Alexis Carrel: Father of Vascular Anastomosis and Organ Transplantation

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Men grow when inspired by high purpose when contemplating vast horizons, sacrifice of oneself is not very difficult for one burning with passion for great adventure

-Alexis Carrel

Alexis Carrel was a French surgeon and biologist, a recipient of the coveted Noble prize in 1912 on his pioneering work on triangulation technique of vascular anastomosis which paved the path for organ transplantation.[1]

Alexis Carrel was born in 1873 in Sainte-Foy-lès-Lyon, France. He was the eldest son of textile manufacturer Alexis Carrel-Billiard and his wife Anne-Marie Ricard. Mother of Alexis carrel undertook embroidering to support her children after demise of Carrel's father due to pneumonia when Alexis Carrel was just 5 years old. Alexis went to a Jesuit day school and college near his home in Lyon. As a schoolboy, he showed an interest in biology by dissecting birds. Alexis was very much impressed by her mother's skill with the tiny needles in embroidery.

He received baccalaureate from St Joseph college in 1900, and subsequently, Doctor of Medicine degree from famous University of Lyon in Europe.[2] On June 24, 1894, French President Sadi Carnot was assassinated by an Italian anarchist in Lyon, which led to portal vein injury and death of Carnot, as surgeons had no means at their disposal to repair artery or veins other than sole ligation. Alexis Carrel has been said to be critical of their faculty for inability to repair the vessel. Carnot's assassination lead to great epiphany in Carrel's mind to develop successful vascular anastomotic skills.

Alexis Carrels' interest to develop fine skills for anastomosis of vessels led him to visit the finest embroiderist in Lyons, Madame Leroidier, to learn use of tiny needles and thread which they employed for fine embroidery work. He developed an extraordinary skill in using the finest needles which he practised on paper many times before starting experiments on animals for vessel anastomosis. In 1898, he joined Anatomical Institute at University of Lyon under celebrated anatomist Jean-Testut (1849–1925), first on human cadavers and later on dogs; Carrel envisaged goal of vascular anastomosis eliminating stenosis, hemorrhage, or thrombosis. In his operating room, he used black towel drapes for surgical field and black gowns to make it easier to see the fine white threads and needles. Smooth-jawed forceps were applied carefully to avoid endothelial injury that could precipitate thrombus formation. Long, straight needles were preferred to short, curved ones [Figure 1]. The fine silk suture was treated with paraffin to minimize its direct contact with the vessel wall. Ringer's solution was used to keep surfaces moisten all the time. A triangulation technique was employed to properly locate the defined margins for the running sutures and then anastomosis was performed by everting the edges of vessel margin to avoid contact of blood other than endothelized smooth surface of vessel wall. Rigid asepsis minimized the risk of infection and facilitated primary healing. In 1902, he published his first manuscript on vascular anastomosis in French literature [Figure 2].[3]

In 1903, Carrel took a pilgrimage to Lourdes and was impressed by the recovery of a young woman with tuberculous peritonitis. He strongly suggested a controlled study of the healings of Lourdes. His account of this experience which reached the lay press had damaging effect on his reputation and surgical career which already was losing momentum due to his critical attitude toward what he considered the antiquated traditions and political sphere of the Lyon medical faculty. Finding a university career blocked by local opposition, he left Lyon and decided to migrate to Montreal, Canada, in 1904.

In Montreal Medical Congress, he has presented a paper which impressed the chairman of physiology from University of

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Figure 1: Alexis Carrel

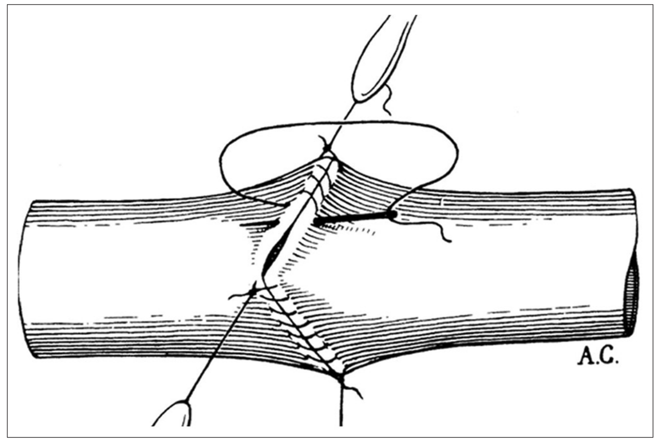

Figure 2: Carrel's technique of vascular anastomosis from the original 1902 publication in Lyon Medical (Courtesy of the Carrel Collection, Georgetown University)

Chicago, who offered him teaching post for which he agreed. In the Hull Physiological Institute between 1904 and 1906, he continued his vascular research with Charles Guthrie. Their productive collaboration resulted in more than 20 publications. Their publications include reproducible techniques of vascular anastomoses, use of vein patches for arterial repair, reimplantation of limbs, transplantation of kidney, heart, ovary, and thyroid xenografts and allografts as well as novel studies of rejection phenomena.[4-6] He also published effective use of venous autografts for replacement of arterial segments and showed it can function as replacement for arteries.[6]

Carrel's growing reputation for surgical skill, bold experimentation, and technical originality gained the attention of a number of leading scientists of that time, including Harvey Cushing, William Halsted, and the pathologist/bacteriologist Simon Flexner, who was the director of the recently established Rockefeller Institute of Medical Research in New York City. A chance meeting between the two during a visit by Carrel to New York led to an invitation to join the Rockefeller Institute. During Carrel's association with Rockefeller Institute from 1906 to 1939, he used cryopreserved vascular homografts to replace the abdominal aorta in cats,[7] tested methods to improve tissue preservation facilitating transplants of carotid arteries,[6] and described the reconstruction of the thoracic aorta by interposition of vena caval grafts and by the use of paraffin tubes as a vascular bypass to prevent spinal cord ischemia.[8] In 1909, he performed intrathoracic operations using tracheal intubation as described by Meltzer and Auer.[8] In collaboration with the French pioneer of thoracic surgery, Theodore Tuffier, Carrel described mitral valvulotomy, annuloplasty, ventricular aneurysmectomy, and coronary artery bypass to the American Surgical Association meeting in 1910.[9-11]

Due to his visionary concepts in surgery during 1904–1906 in Chicago and 1906–1914 in New York, Carrel has been referred to as the "Jules Verne of cardiovascular surgery,"[12] an "innovative wall builder,"[13] a "cardiovascular prophet,"[14] and certainly, the "father of cardiovascular surgery".[15,16]

When World War I began, Carrel was serving his nation French army. He critically noticed that trench warfare that began with the battle of Marne precipitated the frequent occurrence of severe infections, such as gas gangrene. Carrel's interest in wound healing led him to develop the techniques of careful debridement, irrigation, and aseptic dressings to cover wounds.[17] In 1914, Carrel was transferred to the front at The Temporary Hospital, No. 21 in The Rond Royal Hotel at Compiegne. There he established a research laboratory for the study of war wounds through the auspices of The Rockefeller Institute. His collaboration with the English chemist Henry Dakin led to the discovery of the wound irrigant, hypochlorite of soda with bicarbonate buffer ("Dakin's solution"),[18] which has remained in common use even after the advent of antibiotics. Carrel also demonstrated that shock was caused by hypovolemia and best treated by transfusion of blood or washed erythrocytes.[19]

In the 1920s to the mid-1930s, Carrel turned his attention to other subjects, including the etiology of cancer, a field that led him to receive the Nordhoff-Jung Award in 1931, and tissue culture techniques, an area in which he published 124 articles. Interestingly, he maintained a tissue culture experiment of viable chicken heart cells for 34 years.

Carrel later worked with the famous aviator and engineer Charles Augustus Lindbergh (1902–1974) to develop an extracorporeal perfusion pump that was capable of supporting organs *in vitro*. Lindbergh's interest in creating a mechanical heart perfusion pump stemmed from an incident where his wife's older sister contracted rheumatic fever and her doctor said that operation could not be performed as her heart could not be stopped for long enough for surgeons to work on it. They constructed a Pyrex pump in 1935 which sustained a cat thyroid gland for 18 days.[20] This technique formed the groundwork for future developments in heart-lung devices. Between 1935 and 1939, a total of 898 experiments was carried out in Carrel's laboratories using the Lindbergh–Rockefeller Institute perfusion apparatus.[21] In 1938, Carrel also published "The Culture of Organs with Lindbergh" who once remarked "it is astounding to realize how far Carrel was ahead of his time, both in his thoughts and experiments, and how much of his work still remains equal to, or even in advance of, that which has followed."

In 1935, Carrel published his book Man, the Unknown. He put forward theories that humankind could achieve perfection through selective reproduction. Carrel promoted the idea of applying a regime of eugenics ruled by intellectual elite. He believed fascism was superior because it encouraged "men burning with a passion to create." He wrote "Eugenics is indispensable for the perpetuation of the strong. A great race must propagate its best elements." His ideology has met with major criticism and resentment. In 1939, at age 65, he was forced to retire and returned to France. He then became director of the Foundation for the Study of Human Problems set up by the Vichy Government. In August 1943, he suffered a mild heart attack, and he passed away at his home in Paris from heart failure in 1944 following a second more serious heart attack at the age of 71.

It is likely that world may criticize his views of eugenics; but above all, we should acknowledge his great contributions which have saved millions of lives by the development of two major specialties cardiovascular surgery and transplant surgery.

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Conflicts of interest

There are no conflicts of interest.

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